
UNIT 2 SOLVENT EXTRACTION – I

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2.1 INTRODUCTION

In the world of chemicals a substance is rarely found in a pure form. Also whatever may be the method of preparation, a chemical entity, in the truest sense, is never obtained free of impurities. With the rapid growth of chemical technology, the chemists are called upon to devise methods to obtain substances in a pure form. There is another problem that if we look into the magic box of analytical techniques there is hardly a technique which is truly specific for the quantification of a constituent. Both these scenarios necessitate the development of effective methods of separations. The analytical chemists have never lagged behind in developing new methods of separations and bring ingenuity in the existing ones.

In Unit 1 of this course, the general criteria for the selection of a separation technique have been discussed. In a broader sense, the technique to be adopted for affecting the separations depends upon the nature of the host matrix and the associated impurities. Needless to say, the amount to be handled is also important. But whatever may be the situation you always look for a technique which may be convenient, fast and economical. It should be one, which, if the need be, can be conveniently scaled up from the bench level to the plant scale. *Liquid-liquid extraction*, popularly known as *solvent extraction*, comes close to satisfying the above criteria. Besides this, it is a very versatile technique. It is as effective for organic as for inorganic compounds. Moreover, such a broad spectrum of extraction systems is available that any given set of impurities can be dealt with. One definite advantage that the extraction methods offer over the widely used precipitation technique is that much cleaner separation can be achieved by the former. In the precipitation methods, contamination by coprecipitation is a big limitation which is minimized with difficulty, whereas analog of coprecipitation i.e., coextraction (extraction of impurities which should not extract) is almost unknown in solvent extraction. Moreover, precipitation methods fail at micro concentrations because there has to be a sufficient amount of the substance to get precipitated.

Solvent extraction is based on the principle that when a solute is brought in contact with two immiscible solvents, one of which is invariably water and the other organic, the solute distributes itself in a fixed ratio in the two solvents. In certain cases, at some optimum conditions, the solute can be more or less completely transferred from one phase to another. Thus, the technique finds application in separation, purification and enrichment. There is another important aspect of solvent extraction in the form of extractive spectrophotometry. The absorbance of coloured metal complexes, particularly metal chelates, extracted in the organic phase is measured. In favourable circumstances, it may be possible to concentrate the metal, separate it from interferences and develop the absorbing system in a single step. Thus, the extractive spectrophotometry invariably scores over normal spectrophotometry in terms of selectivity and sensitivity.

It may be important to point out here that the general principles, terminology, and methodology used for the separation of inorganic and organics are the same. The scope of applications of the technique for the two types of materials is by no means different. But in order to illustrate the principles more vividly and the variety of situations that are encountered, our focus will be mainly on the extraction of metal ions. The different aspects of the technique have been dealt with in two units, the present one and Unit 3.

The next unit on solvent extraction will deal mainly with different types of extraction equilibria, factors influencing extraction, different approaches for metal ion separations and the criteria for the choice of organic phase.

Objectives

After studying this unit, you should be able

- explain general principles and terminology used to express partitioning of solute and efficiency of separations,
- classify extractants used for metal ions, and
- the role of diluents and modifiers in the extraction of metal ions.

2.2 GENERAL PRINCIPLES AND TERMINOLOGY

This particular section focuses on a system where a solute is brought in contact with two immiscible solvents. What will be the constraints limiting the partition of the solute in the two phases? How will be the distribution quantified? Can the distribution pattern of the solute be evolved into a method of separation or purification of the solute? Some basic relevant terms are being discussed here.

2.2.1 Phase Rule Consideration

Let us consider the situation when a solute is in contact with two immiscible solvents. If we apply Gibbs Phase Rule to this system,

$$P + V = C + 2$$

where P is the number of phases, V is the variance or degrees of freedom and C is the number of components.

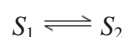
Here, we are dealing with two phases, i.e. two immiscible solvents and one solute distributed between them. Therefore, variance will be unity. This will amount to the fact that the concentration of the solute in one phase will fix the concentration of the solute in another phase. There will be a definite relationship between the solute

concentration in each of the two solvent phases. This ultimately forms the basis of the *distribution law*.

2.2.2 Distribution Law and its Limitations

It is clear from the above discussion that if the system is composed of two immiscible solvents and one distributing solute, there is one degree of freedom. The ratio of the concentration of the solute in the two phases is invariant. It is independent of the total concentration. Based on this, the distribution law was first enunciated in 1872 by Berthelot and Jungfleish and later elaborated by Nernst in 1891. The law popularly known as a **Nernst distribution** law puts greater restraints. It can be stated as that *a solute will distribute itself between two immiscible solvents in such a manner that, at equilibrium, the ratio of concentration of the solute in the two phases at a particular temperature will be a constant provided the solute has the same molecular weight in each of the phases.*

For a solute S distributing between solvents 1 and 2, we can write



$$K_D = \frac{[S]_2}{[S]_1}$$

where K_D is known as the *distribution coefficient* or the *extraction coefficient*. It is independent of the total concentration of the solute and the phase volumes.

Hence, $[S]_1$ and $[S]_2$ denote the concentrations of the solute in phases 1 and 2, respectively. If you imagine these total concentrations to be increased until one phase is saturated and reaches the concentration $[sS]_1$, the second phase must simultaneously become a saturated solution of concentration $[sS]_2$. The distribution ratio, $[sS]_2 / [sS]_1$ then becomes the ratio of the solubilities of the solute in the two phases. This expression holds reasonably well for materials which are extremely sparingly soluble.

The Nernst distribution law is a useful approximation for the distribution of a species in the two phases. Careful experiments reveal two serious shortcomings. The first shortcoming is encountered when the distributing species is involved in chemical reaction such as dissociation or association and is changed in either phase.

A very typical case in this regard is the distribution of benzoic acid between water and benzene. In the aqueous phase, benzoic acid is partly ionized while in benzene it is partly dimerized by hydrogen bonding in the carboxylic group. The pertinent equilibria dealing with this case will be discussed in the next sub-section.

In the extraction of metal ions, the existence of a single species can be properly accounted for while considering the distribution of the species in the two phases. They definitely add complexity to the distribution expression but this does not represent any essential inadequacy of the distribution law itself.

The second shortcoming arises because Nernst law as stated above is not thermodynamically rigorous. Strictly in the thermodynamic terms, it is the activity ratio rather than concentration ratio that should be constant. In other words, the thermodynamically rigorous partition law would be stated as that the activity of a chemical species in one phase maintains a constant ratio to the activity in the other phase. This precisely defined ratio is known as *thermodynamic partition coefficient* or *distribution coefficient* (K').

$$K' = \frac{\{S\}_2}{\{S\}_1} = \frac{\gamma_2[S]_2}{\gamma_1[S]_1} = \frac{\gamma_2}{\gamma_1} K_D$$

$$\text{or } K_D = \frac{\gamma_1}{\gamma_2} K'$$

where $\{S\}$ represents the activity of the solute S and γ is its activity coefficient.

The Variation in the distribution coefficient K_D can be seen due to variation in the activity coefficients in each of the phases. So long the solute concentration is very low and the activity coefficients approach unity, the value of K_D is constant. The use of radiotracers for distribution studies can be very helpful to investigate the range of concentration in which the activity coefficients approach unity.

By using radiotracers, the concentration can be kept very low and the distribution of the species in the two phases can be very accurately and conveniently quantified. Using radiochemical technique, it has been observed that the distribution coefficient of gallium chloride between ethyl ether and 6 M HCl remains constant (within 5%) over a concentration range $10^{-12} - 2 \times 10^{-3}$ M GaCl_3 . At higher concentration ranges, constant values of distribution coefficient might be expected over ranges in which the ratio of the activity coefficients remains constant. Parallel variations in the activity coefficients over extensive concentration ranges in two solvents are not likely to occur.

2.2.3 Distribution Ratio

One particular aspect that should be amply clear to you that the distribution of a species in the two competing phases is not governed by a simple equilibrium. The situation is complicated by the chemical interactions in the two phases and they can eventually markedly affect the concentration of the distribution species. In this context, you have already been told the case of distribution of benzoic acid between water and benzene.

It is rarely observed that a single species may be present in the two phases. But as an analytical chemist, for the purposes of separations, you are concerned with the overall or stoichiometric distribution of the component of interest in the two phases. In whatever chemical forms the solute remains in the two phases, it is immaterial. Also, it is easier to determine the overall distribution of the component in the two phases. Therefore, to describe the extraction there is a more practical quantity known as "Distribution Ratio" (D).

$$D = \frac{\text{Total concentration of } S \text{ in organic phase}}{\text{Total concentration of } S \text{ in aqueous phase}}$$

In an ideal condition, if there is no interaction in either of the phases, D would be the same as K_D . If all the interactions of the distributing species are known, they can be evaluated and an expression showing the effect of various parameters on D can be obtained. In order to illustrate the point, we discuss two equilibria one from organic and the other from inorganic chemistry.

A. Distribution of Benzoic Acid Between Benzene and Water

In the aqueous phase, benzoic acid is partially ionized



In the benzene phase, partial dimerization of benzoic acid takes place



The three species, HBz, Bz⁻, (HBz)₂ will have their own K_D values

$$K_{D_{HBz}} = \frac{[HBz]_o}{[HBz]_a} \quad \dots (2.3)$$

$$K_{D_{Bz^-}} = \frac{[Bz^-]_o}{[Bz^-]_a} \quad \dots (2.4)$$

$$K_{D_{(HBz)_2}} = \frac{[(HBz)_2]_o}{[(HBz)_2]_a} \quad \dots (2.5)$$

Subscripts “o” and “a” represent the organic and aqueous phases. It is known that the benzoate ion remains almost completely in the aqueous phase and the benzoic acid dimer in the organic phase. As a chemist, you will usually be interested in knowing where is benzoic acid, whether a part of this acid is ionized or dimerized.

$$D = \frac{\text{Total benzoic in the aqueous phase}}{\text{Total benzoic in the organic phase}} \quad \dots (2.6)$$

$$D = \frac{[HBz]_o + [(HBz)_2]_o}{[HBz]_a + [Bz^-]_a} \quad \dots (2.7)$$

It is evident that D will not remain constant over a range of experimental conditions. Suppose we increase the pH of the aqueous phase, D will be lowered. This is because benzoic acid will be converted to benzoate ion which does not extract in the organic phase.

B. Distribution of Osmium Tetraoxide between Carbon Tetrachloride and Water

In the aqueous phase, osmium tetraoxide exists as a dibasic acid, H₂OsO₅, which partially dissociates as shown below:



In the organic phase, osmium tetraoxide forms a tetramer



If we analyse the total scenario with regard to the presence of osmium in its different forms in the two phases, it is not simple. Osmium tetraoxide monomer will be the only species present in both the phases. H₂OsO₅ and the charged species, HOsO₅⁻, OsO₅²⁻ will remain totally in the aqueous phase while (OsO₄)₄, the tetramer, will remain in the organic phase.

$$D = \frac{\text{Total concentration of osmium in the organic phase}}{\text{Total concentration of osmium in aqueous phase}} \quad \dots (2.12)$$

$$D = \frac{[OsO_4]_o + 4[(OsO_4)_4]_o}{[OsO_4]_a + [H_2OsO_5]_a + [HOsO_5^-]_a + [OsO_5^{2-}]_a} \quad \dots (2.13)$$

The value of D or the total osmium concentration in the two phases can be calculated if the values of equilibrium constants of the reactions listed above are known.

In the two equilibria, one of benzoic acid and the other of osmium tetroxide, the distribution ratios have been presented in terms of concentration of certain species. The expressions can be simplified by bringing in certain equilibrium constants and minimum number of variables. Finally, the effect of variables on distribution ratio can be experimentally verified. For reference, see answer of Terminal Question 1.

2.2.4 Percentage Extraction

Distribution ratio (D) is merely a concentration ratio and the extent of extraction will, therefore, depend upon the volume ratio of organic to water phases. Since we are mainly interested in the extent of extraction, we go for another practical term which is known as percent extraction ($\% E$) or E .

Let f be the fraction of the total metal in the organic solvent at equilibrium, then $(1-f)$ is its fraction in the aqueous phase. Let V_o and V_a be the volumes of the organic and aqueous phases, respectively. Then

$$D = \frac{\text{Total metal in organic phase}}{\text{Total metal in aqueous phase}}$$

$$= \frac{[M]_o}{[M]_a} \quad \dots (2.14)$$

$$D = \frac{fV_o}{1-fV_o} = \frac{fV_a}{(1-f)V_o} = \frac{fV_a}{V_o - fV_o} \quad \dots (2.15)$$

$$DV_o - DfV_o = fV_a \quad \dots (2.16)$$

$$DV_o = DfV_o + fV_a \quad \dots (2.17)$$

$$DV_o = f(DV_o + V_a) \quad \dots (2.18)$$

$$f = \frac{DV_o}{V_a + DV_o} \quad \dots (2.19)$$

$$\text{Percentage of metal extracted} = 100f = \frac{100D}{D + (V_a/V_o)} \quad \dots (2.20)$$

If the volumes of the two phases are equal, then the denominator simplifies to $D + 1$. *i.e.*,

$$\% E \text{ or } E = \frac{100D}{D + 1} \quad \dots (2.21)$$

The percent extraction is a more useful term for reporting the analytical data. Fig. 2.1 shows the interconversion of D and $\% E$ at a unit volume ratio. It can be seen from the plot that when the extraction efficiency approaches 100%, the distribution ratio approaches infinity as a limit. For the extractions to be in the range of 99 – 100% the distribution ratio will vary from 99 to infinity. Such a large difference in the distribution ratio is 99 and infinity may be misleading when in reality the extraction would be nearly complete in both the cases.

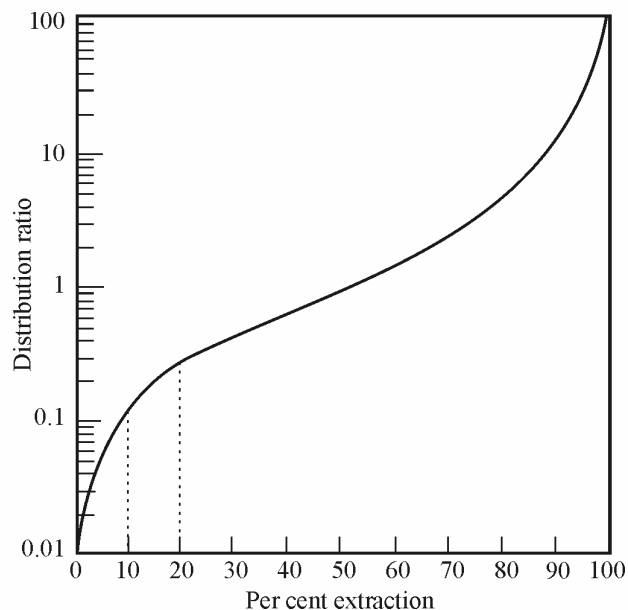


Fig. 2.1: Relation between distribution ratio and percent extraction

2.2.5 Separation Factor

Solvent extraction is basically a separation technique. The main objective is to separate the desired constituent from the other substances present in the mixture. In order to achieve the objective, we look for a solvent extraction system which may quantitatively extract the required constituent with practically negligible amount of the impurities present. The other way to solve the problem is to quantitatively extract all the impurities leaving behind the desired constituent totally in the aqueous phase. The situation where none of the impurities or all the impurities may quantitatively transfer from one phase to the other is rarely encountered. We are likely to face situations where they differ only somewhat in their tendencies to pass from one solvent to another. In order to achieve a simple separation, it is necessary that the distribution ratio of the material of interest and the interference be sufficiently different. The effectiveness of separation is usually expressed by means of separation factor “ β ” which is related to the individual distribution ratios in the following way:

$$\beta = \frac{(C_1)_o / (C_2)_o}{(C_1)_a / (C_2)_a} \quad \dots (2.22)$$

$$= \frac{(C_1)_o / (C_1)_a}{(C_2)_o / (C_2)_a} \quad \dots (2.23)$$

$$= \frac{D_1}{D_2} \quad \dots (2.24)$$

C_1 is the concentration of component 1 in the organic and aqueous phases further labeled by subscripts o and a, respectively and C_2 is the concentration of component 2 in the two phases. D_1 and D_2 are the distribution ratios of components 1 and 2, respectively. For a good separation, β should deviate from unity as largely as possible. The ingenuity and knowledge of a separation chemist is tested to find conditions where β should deviate from unity. With the available data and knowledge of chemistry, in most of the cases, it is not a problem. An elaboration of this will be taken up later in Unit 3, Sec. 3.4 where a detailed discussion on the different approaches adopted for metal ion separation are included. Although most of the separations involving solvent extraction are based on favourable separation

factors, it is occasionally necessary to resort to separations where the distribution ratios of the components of the mixture may not be much different. In such situations, it is necessary to employ fractionation methods in which distribution, transfer and recombination of different fraction is advised to be performed a number of times to achieve a certain level of separation.

2.2.6 Multiple Extractions

It should be clear in mind that one transfer does not necessarily lead to a clean separation. Under these situations, it is better to combine a number of successive partial separations until the desired level of purity is achieved. While considering how two phases may be brought together repetitively, the following four types of situations may be visualized.

- i) It is simple where one shot contact between the two phases is made. The two phases are shaken until an equilibrium is attained.
- ii) In this, one phase could be brought repeatedly in contact with fresh portions of a second phase. This situation would be applicable where one substance remained quantitatively in one phase while the other one was distributed in the two phases. This would be like the repeated extraction of an aqueous solution with successive portions of an organic solvent. The example of Soxhlet extractor will fall in this category. Another analogy may be provided by reprecipitation in gravimetric analysis.
- iii) In this case, one phase moves while in contact with the second phase which remains stationary. Like the various chromatographic techniques the mobile phase moves. This is sometimes designated as countercurrent. But in the real sense, it is not countercurrent as only one phase moves. Therefore, it is actually known as *discontinuous countercurrent* or *pseudo countercurrent*.
- iv) This case is really countercurrent in which both phases move continuously in contact with each other in *opposite directions*. A true example of countercurrent is *fractional distillation*.

2.2.7 Multiple Extractions with Successive Portions

We have seen the importance of batch extraction because it is most widely used for analytical separations. In this procedure, a given volume of solution is brought in contact with a solvent until equilibrium is attained. Even if we suppose that none of the interfering substances is extracted, the completeness of the extraction is a very important factor. It has already been pointed out that the distribution ratio is a concentration ratio and, therefore, the actual fraction of the solute extracted will depend upon the ratio of volumes of the two solvents competing for the solute.

Let v_1 mL of solution (phase 1) containing w g of solute is contacted to equilibrium with v_2 mL of another solvent (phase 2) immiscible with the first. Let w_1 is weight of the solute remaining in the phase 1 after equilibrium is attained.

Now concentration in phase 1 (C_1) = w_1/v_1 g mL⁻¹
 And concentration in phase 2 (C_2) = $w-w_1/v_2$ g mL⁻¹

$$\text{Therefore, } D = \frac{w-w_1/v_2}{w_1/v_1} \quad \dots (2.25)$$

$$\text{And, } w_1 = w \left(\frac{v_1}{Dv_2 + v_1} \right) \quad \dots (2.26)$$

If we again extract phase 1 with another portion v_2 mL of solvent, w_2 g will be the weight of the solute remaining in the phase 1. After this extraction,

$$w_2 = w_1 \left(\frac{v_1}{Dv_2 + v_1} \right) \quad \dots (2.27)$$

Therefore,
$$w_2 = w \left(\frac{v_1}{Dv_2 + v_1} \right)^2 \quad \dots (2.28)$$

If the same volume of the extracting solvent is used for each successive extraction and w_n g is the weight of the solute remaining in phase 1, after n extractions w_n will be given by the expression

$$w_n = w \left(\frac{v_1}{Dv_2 + v_1} \right)^n \quad \dots (2.29)$$

The fraction remaining in phase 1 after n extractions $= f_1 = \frac{w_n}{w} = \left(\frac{v_1}{Dv_2 + v_1} \right)^n \quad \dots (2.30)$

And, the fraction transferred to phase 2 $= f_2 = 1 - f_1$

If phase 1 is aqueous and phase 2 is organic, then

$$\% \text{ Extracted} = 100 f_2$$

For the most effective extraction, v_2 should be taken as small as possible and n as great as possible for the given amount of a solvent. This amounts to the fact that the best results are obtained by a relatively large number of extractions with small amounts of solvent. The fact is illustrated in the answer to Terminal Question 2.

SAQ 1

What will be the conditions when distribution coefficient, thermodynamic distribution coefficient and distribution ratio are the same?

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SAQ 2

The partition coefficient of iodine between chloroform and water is 132. What percentage of iodine dissolved in 100 mL of water will remain in the aqueous phase if it is contacted to equilibrium with 50 mL of CHCl_3 ?

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Two metals A and B form chelates with dithizone and extracted with chloroform. What should be the ratio of their distribution ratios so that 99% of B remains in the aqueous phase and 99% of A goes to the organic? Assume equal volumes of organic and aqueous phases.

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2.3 CLASSIFICATION OF EXTRACTION SYSTEMS

It should be now clear to you that the extraction basically takes place in the following three steps:

- i) Formation of an uncharged complex or extractable species.
- ii) Distribution of the extractable complex or the species.
- iii) Interaction of the complex or the species with the organic phase.

These three aspects are shared by almost all extractions and serve as a basis of a useful organisational pattern.

In order to discuss in a systematic manner the various extraction systems used for the extraction of different metal ions, we will have to classify them into certain groups based on different approaches. Several classification schemes based on different approaches have been proposed. Such schemes are generally based on the type of extractants or the chemical reactions involved in the extraction process.

It may be important to point out here that the list of metal ion extractants is very large. They range from simple to complex molecules and single compound to mixtures. One or the other aspect of these extractants is more or less thoroughly investigated. From time to time, various types of extractants with variegated modes of extractions have been synthesized and investigated. Some of them have desirable attributes and are able to permeate in the market for commercial applications. It is a known fact that the use of a majority of extractants is confined up to the bench level and cannot be extended to plant scale. Our discussion in this section will cover the gamut of metal ion extractant but the focus will be on the commercial products.

With this background, a logical consequence seems to be that whatever may be the criteria for classification it may not yield a universal system. However, a simpler classification incorporating the mechanism of extraction and the extractant used into different types, is given as under

1. Distribution of simple molecules.
2. Extraction by compound formation.
3. Extraction by salvation.
4. Extraction by ion pair formation.

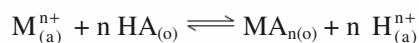
A special mention is being made of crown ethers as extractants.

2.3.1 Distribution of Simple Molecules

This involves the distribution of simple covalent molecules like GeCl_4 , HgBr_2 and AsBr_3 between aqueous phase and inert organic solvent.

2.3.2 Extraction by Compound Formation

The reagents which fall under this category are called *liquid cation exchangers*. These extractants operate, at least, formally by the exchange of the H^+ ions of the acidic organic compound with the cation present in the aqueous phase



The process is similar to solid-liquid ion exchange in which HA is equivalent to an ion exchange resin. Although there is some similarity between the reagents of this class and solid ion exchangers, very important differences exist as regards their interaction with the metal ions. These cation exchange extraction systems can be broadly classified under the following three categories.

- i) Extraction by chelating agents.
- ii) Extraction by acidic organophosphorus esters.
- iii) Extraction by carboxylic and sulphonic acids.

There are some important differences between extraction processes involving chelates on one side and phosphorus and carboxylic acids on the other. The acids forming chelates are usually weak and more soluble in organic than in inorganic solvents. The composition of the chelates is usually unaffected by the nature of the diluents. The extraction mechanism in these cases is well described by simple equation of Law of Mass Action. In several classification systems, the chelates forming system, forms a separate class by itself.

If we look at the extraction equilibria involving carboxylic and phosphorus acids, they are not easily described by the distribution law equations. The acids are stronger than those forming chelates. Weak complexes are formed and a high ligand to metal ratio is required for their formation. They are more water soluble. Extraction of metals by organophosphorus, carboxylic and sulphonic acids is affected more by solvent phase properties than it is for chelating agents. Both organophosphorus and carboxylic acids often form dimers or polymers in the organic phase.

After having discussed the salient differences in the extractants within this category, it is essential to look into the characteristic features of their interaction and the important ones available for metal ion extraction.

i) Chelating Agents

The chelating agent acts as a weak acid and contains a donor group so as to form a bidentate chelate with the metal. The reagent invariably neutralizes the charge on the metal ion and satisfies the coordination number requirement. If the coordination sites remain vacant, they may be occupied by the solvating molecules which sometimes may be ligand itself. Hydrated metal chelates are poorly extractable in non polar solvents which do not have the tendency to replace the water of hydration.

The above discussion amounts to the fact that the chelating agents which form uncharged metal complexes with the metal ion have at least two functional groups. One of these is either OH or SH not necessarily attached to the carbon atom. The other is a basic functional group capable of coordination by its donor properties. The stable ring formed is usually five or six membered. The stability of the chelate will depend on a number of factors namely basic nature of the

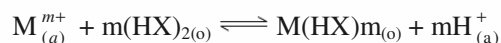
coordinating site, the acidity of the OH/ SH group, coordination number and coordination ability of the metal. Both the acidity of the OH/ SH group and the basic strength of the donor atom is affected by the structural factors. Also, the relative position of the active sites in the molecule will affect the extractive effectiveness of the molecule by determining the size of the ring.

The reagents included under this category are weak or very weak acids, therefore the formation of extractable complexes will much depend upon the pH of the aqueous phase. Adjustment of pH of the aqueous is a useful way for the extractive separation of metal ions if the equilibrium constants of the metal ions in question are sufficiently different. It is possible to achieve selectivity by factors other than pH control. One of the most popular approaches is by use of *masking agents* which form water soluble complexes of higher stability than those formed by the chelating agents.

The extractability of the metal chelate is affected by its solubility characteristics in organic solvents. If a methyl group on the molecule of a chelating agents is substituted by a benzene ring, an increased solubility of metal chelate in non-polar solvents is observed. Generally, introduction of a hydrophobic group in the chelating molecule favours the transfer of its metal chelate in the organic phase. If the coordination number requirements of the metal in the metal chelate are satisfied by the water molecules, the chelate is poorly extracted in non-polar solvents. An increased extraction of these chelates is observed by oxygen containing diluents, polar in nature such as ketones, esters and alcohols. This increased extraction of metal chelates is attributed to their ability to replace water molecule in the metal chelate. Such a cooperative effect of a chelating agent and a solvent with solvating properties to enhance the extraction of metal will be dealt under the head of “synergistic extraction”. The well-known reagents under the head of chelating agents are β -diketones, dithizone, 8-hydroxyquinoline, monoximes, dioximes, nitrosophenols, nitrosoarylhydroxylamines, hydroxamic acids and dithiocarbamates. Some of the acidic chelating agents commercially available in the market for the hydrometallurgical operations are LIX63, LIX64N, LIX65N, LIX70, SME-529, P-5000 series (hydroxyoximes); Kelex 100 (oxine derivative) and Hostarex DK16, LIX 54, X 151 (β -diketones).

ii) Acidic Organophosphorus Compounds

Alkylphosphoric, phosphonic and phosphinic acids fall under this category. They also extract the metals by a cation exchange reaction. As mentioned earlier, the experimental data of these extractions do not necessarily fit into the mass equation of exchange equilibrium. The organophosphorus acids have pronounced tendency of association into dimers and larger aggregates. The other problem arises due to solvation of the metal extractant complex by one more extractant. The extraction of metal by dimers of the extractant can be represented as



This will mean that the extraction is proportional to the m -th power of dimer concentration and inversely proportional to the m -th power of the proton concentration, where m is the charge on the metal ion.

The type of alkyl or aryl group of the organophosphates will influence the acidic nature of the reagent. Increase in the chain length or branching of the substituent will decrease the acid strength. Although the main mode of action of these extractants is that of cation or hydrogen exchange, there are numerous

cases where the coordination of the phosphoryl oxygen occurs. This results in the formation of chelation products mainly polymers. Under these circumstances, the substituents on the phosphorus atom play an important part which frequently, is contrary to their role in hydrogen exchange mechanism.

It is possible to make certain generalizations regarding the extraction of metals as a function of the properties of the metal ion. As the charge on the metal ion increases, then so does its extractability. For metals having the same charge, extraction varies inversely as the ionic radius. This means small ions extract better than large ions having the same charge. Thus, in general, at a fixed pH , the monovalent alkali metal exhibits the lowest extractability and ions like Ce^{4+} and Th^{4+} exhibit much higher extractability. This is the reason that sodium or ammonium salt of DEHPA (di-2-ethylhexyl phosphoric acid) can be used for the extraction of metal ions having charges greater than +1. The type of anionic species present in the aqueous phase also affect the extraction.

The nature of organic diluent *vis-a-vis* the polarity and dielectric constant, is reported to affect the extraction. The effect is certainly complex, depending to an unknown extent, on the dielectric constant, dipole moment and polarizability of the diluent and the complex and the hydrogen bond donor and acceptor abilities. It has been shown that the extent of polymerization of the extractants is related to its interaction with the diluent. And, this in turn affects the extraction of the metallic species. The effect of the diluent may be drastic enough to affect the composition of the organic phase complex. In diluents where the extractant is predominantly monomeric, the composition of the complex is invariably MX_m rather than $M(X\ HX)_m$.

The development of this class of extractants, at the initial stage, was closely associated with the objective of finding organic phosphorus compounds for practical solvent extraction applications in the field of processing of nuclear fuel. At an early stage of the use of tributyl phosphate (TBP) for nuclear fuel processing, it was recognized that the acid hydrolysis products of this reagent—the mono and dibutyl phosphoric acids, were themselves very effective extractants. This observation prompted a systematic survey of alkyl phosphorus compounds for metal extraction chemistry. The list of this class of extractants is very big and it may not be possible to mention all of them. Of all the mono esters, mono-2-ethylhexyl phosphoric acid is the most commonly used extractant and among the dialkylphosphates, di-2-ethylhexyl phosphoric acid (DEHPA) has by far received the maximum attention. Besides these two, a number of them like octylphenyl phosphoric acid (OPPA), diisodecyl phosphoric acid (DP-10R), 2-ethylhexyl phosphonic acid (PC-88A), 2-ethylbenzyl phosphonic acid, dibutyl butylphosphonate (PC-44) and *bis* (2,4,4-trimethylpentyl) phosphinic acid (Cyanex 272) are well known. In the nineteen eighties, American Cyanamid, marketed the mono and disulphur analogues of Cyanex 272, *i.e.*, *bis* (2,4,4-trimethylpentyl) monothiophosphinic acid (Cyanex 302) and *bis* (2,4,4-trimethylpentyl) dithiophosphonic acid (Cyanex 301). The introduction of sulphur in these acids lowers the pK_a of the acids and it may be possible to work with them and extract at a lower pH .

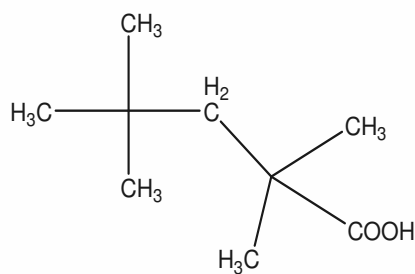
iii) Carboxylic and Sulphonic Acids

The solutions of carboxylic and sulphonic acids, in non-polar organic solvents, are similar to cross linked resinous cation exchangers. They frequently offer some advantages and often only require a pH adjustment of the aqueous phase for achieving the separation of elements similar in properties. Different carboxylic, alkyl or aryl-alkyl sulphonic and sulphuric acids form salts with a number of metal ions including alkali and alkaline earths. These salts are

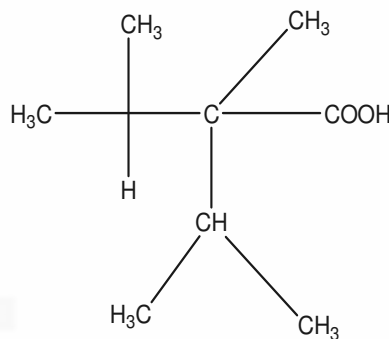
frequently less soluble in aqueous solutions but more in some organic solvents. The extractability of these acids generally parallels their basicity.

Solvent Extraction-I

The metal extraction by these extractants is complicated by their state of aggregation in solution. The polymerization of these acids, although, does not go beyond dimers but the extensive aggregation of their salts plays a significant role in their extraction. The bulk of available data on the extraction by carboxylic acids is on normal or branched chain mono carboxylic acids with seven or more carbon atoms in the chain. Under certain specified conditions, the aliphatic and aromatic hydroxyl carboxylic and dicarboxylic acids can also extract certain metal ions. Various benzoic acid derivatives, dinitrobenzoic, cinnamic and phenylacetic acids and amino and perfluoro derivatives of *n*-carboxylic acids have also been used as extractants. Out of commercial extractants of this class, Versatic and Naphthenic acids obtained from the distillation of crude petroleum are important.

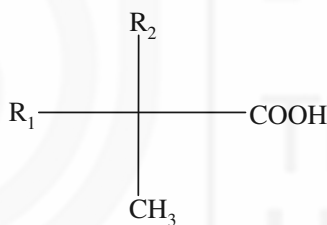


(56%)



(27%)

Versatic 9

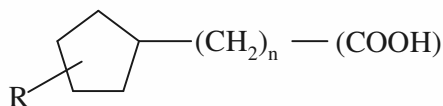


(R₁, R₂, = C₄ - C₅)

Versatic 9-11

Other Versatic acids having a similar structure are Versatic 10, 13 and SRS-100.

Naphthenic acid is the name given to a group of cyclic aliphatic monocarboxylic acids having the following general structure:



Naphthenic acids

This fraction of crude petroleum has a molecular weight in the range of about 170 – 330.

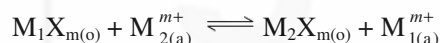
Aliphatic and aromatic sulphonic acids are represented by the formula RSO_2OH (R = straight/ branched chain or aromatic saturated radical). A variety of these extractants are available and more useful of these reagents are dodecylbenzene and dinonylnaphthelene sulphonic acids. Aliphatic sulphuric acids with the general formula ROSO_3H (R = alkyl chain usually at least 15 carbon atoms), have been shown to be potentially useful extractants.

Carboxylic, naphthenic and sulphonic acids are associated in organic solvents. This is based on considerations of medium effect on the extent and strength of inter molecular hydrogen bonding in such compounds. The effects of aqueous phase parameters on the extraction of metals are similar in all the different type of acids. The major difference between the sulphonic (and sulphuric) and carboxylic acid extractants is the capacity of the former to extract cations from solutions below pH 1. As a rule, the distribution ratio decreases with increasing acidity.

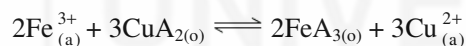
Dimerization of carboxylic acids in non-polar solvents is well established. The extracted species are invariably solvated by one or more extracted molecules. Carboxylic acids also have a tendency to higher polymerization with increase in concentration of extracted metal. In the extraction of Ni(II) , Co(II) and Cu(II) by naphthenic acid, the extracted species have been reported as $\text{Ni}_2 \cdot \text{A}_4 \cdot 4\text{HA}$, $\text{Co}_2\text{A}_4 \cdot 4\text{HA}$, $\text{Cu}_2\text{A}_4 \cdot 4\text{HA}$, respectively.

The composition of the aqueous phase to a considerable extent affects the extraction of metal by carboxylic and sulphonic acids. As far as pH is concerned, there is no qualitative difference. Metals are preferentially extracted from aqueous solutions containing alkali. Sometimes, there is no extraction in its absence. An alternative way to increase the distribution is to use sodium salt of fatty acid instead of free acids.

Carboxylic acids react with a number of metal ions and the extraction can be made selective by adjusting the pH . Moreover, in the metal carboxylate extraction, the effect of electronegativity is so prominent that a cation exchange takes place.



The less electronegative element sodium is quantitatively exchanged in the organic phase by any more electronegative element say cobalt. Cobalt can be exchanged by Cu and Cu for Fe. Naphthenic acid extracts metal with increasing pH in the order $\text{Fe}^{3+} < \text{Cu}^{2+} < \text{In}^{2+} < \text{Co}^{2+} < \text{Fe}^{2+} < \text{Mn}^{2+}$ over pH 1-10. Thus, a metal in the extraction series can be displaced from the organic phase by a metal to its left in that series.



The use of carboxylic acid as metal extractants for commercial purposes is limited because of extractants loss, emulsion formation tendency and slow kinetics of the extraction.

2.3.3 Extraction by Solvation

An important group of solvent extraction systems is based on the solvation of neutral inorganic molecules or complexes by electron donor containing extractants. Solvation increases the solubility of inorganic species in the organic phase. The oxygenated solvents like ethers, esters, alcohols and ketones and organophosphorus esters solvate the proton and neutral inorganic species making the extraction of the acids and salts feasible. The simple examples are $\text{Ce(NO}_3)_4 \cdot (\text{ether})_2$, $\text{HNO}_3 \cdot \text{TBP}$, $\text{VO}_2(\text{NO}_3) \cdot (\text{TBP})_2$ and $\text{H}^+ (\text{TBP})_4 \cdot \text{ReO}_4^-$ where TBP is tri-*n*-butyl phosphate.

The large variety of solvating molecules can be grouped into two distinct categories as follows:

- i) Reagents containing oxygen bonded to carbon such as ethers, esters, alcohols and ketones.
- ii) Reagents containing oxygen or sulphur bonded to phosphorus e.g. alkylphosphates or alkylthiophosphates.

Despite the similarity between the two categories, several features emphasize the differences. It is the strong polar character of compounds of category (ii) which is responsible for the differences.

A major difference between these extractants types is the role of water. Strongly polar organophosphorus compounds compete favourably with water and can replace water molecules in the inner coordination sphere of the metal ion. With ethers and ketones, water is a necessary part of the complex. Take the example of extraction of uranyl nitrate, it is transferred to dibutyl carbitol with four molecules of water while it goes all alone in TBP. Moreover, when phosphorus compounds are used as extractants, it is easier to evaluate the degree of solvation while in the case of class (i) extractants, it is difficult. In the latter case really, a number of mixed hydrates – solvates coexist in equilibrium. In this regard a typical case, worth mentioning is the variation in the water molecules in the extracted species when iron (iii) chloride is extracted in ether.

With the above background, we should now look into important specific characteristics of the two categories of solvating extractant.

1. Extractants Containing Carbon-Oxygen Bonds

As mentioned earlier, the extractants of this class are essentially ethers, ketones and alcohols. They are electron donating reagents but alcohol being amphoteric, exhibits both donor and acceptor properties. These extractants did receive attention during the earlier part of development of metal extraction chemistry. Diethyl ether and diisopropyl ether were at one time very favoured extractants for metal halides and halometallic complexes. They were also used as extractants for metal nitrates resulting into the development of separation technology for nuclear fuel reprocessing. Metal sulphates and perchlorates are generally nonextractable into these solvents. Only one ketone, methyl isobutyl ketone (MIBK) has been used commercially for the extraction of metals.

It has been shown that inorganic salts, soluble in water immiscible oxygen bearing organic solvents, are usually weak electrolytes or they involve a complex ion. The main characteristic of extraction from aqueous halide solutions by oxygen containing solvents is the extraction of protonated metal complex anions, preferentially singly charged ones, ($H^+.FeCl_4^-$). An equally important feature of the system is the high extractability of essentially covalent metal halides. The striking difference between the two is that the halo complexes are extractable into oxygenated solvents only while the halides can be transferred into nonpolar hydrocarbons also. With some restrictions due to aqueous complex chemistry of metals in halide solutions, higher extractions are observed for monovalent complex anions of bivalent and trivalent transition metals. In spite of pronounced complexing tendencies of bivalent metals, the high charge on the complex MX_4^{2-} prevents their extraction into relatively high dielectric constant solvents such as butanol and octanol.

As regards the extraction of metal nitrates, a particular aspect has received special attention *i.e.*, the role of salting out agents. It has been observed that the addition of nitrate salts to the aqueous phase greatly enhances the extraction. The extraction of uranyl and thorium nitrate in ether is greatly enhanced by the addition of ammonium nitrate. It was found that other metal nitrates could be

used for the purpose but the multivalent nitrates with pronounced hydration tendencies proved to be the most effective. The function of the salting out agents is considered to be three-fold as is given below:

- i) Providing a higher concentration of complexing anion which by the *Law of Mass Action*, would increase the concentration of the complex and thus, improve the extraction.
- ii) Water will bind to the ions of salting out agent and thus, become unavailable as free solvent to compete with the organic phase for partitioning.
- iii) The addition of salt will lower the dielectric constant of the aqueous phase which will favour the formation of ion association complex. It is the neutral ion association complex which is getting extracted.

If one looks at the organic phase characteristics, it has been observed that the extraction power changes with the class and solvent molecule. In other words, the solvent efficiency remains a specific characteristic of any particular system. For example, generally ethers are better extractants than ketones for hexavalent actinide nitrates while trend is opposite for tetravalent metal halides from aqueous solutions. Within a homologous series, there is a lowering of metal extraction on increasing the hydrocarbon character of the organic solvent. This means that the lengthening of alkyl chain in alcohols, ketones and ethers by introducing methylene group, will cause a decrease in the extractive power. The dipole moments of ethers and ketones are affected by the symmetry of the molecules. As a rule, asymmetric ethers and ketones have a greater efficiency than the symmetric ones. The greater dipole of asymmetric extractants leads to an increased electron density at the donor oxygen and thus, increased tendency to coordinate. The distribution ratio of metal decreases with increasing branching of the alkyl group, the effect is particularly marked when the branching is near the active oxygen.

As of today, the phosphorus bonded oxygen donor extractants have completely phased out the ethers and ketones from technological applications.

2. Extractants Containing Phosphorus-Oxygen Bonds

Extractants of this group could be considered as derivatives of phosphoric acid. The general structures of the esters of organophosphorus acids are given in Table 2.1.

Table 2.1: General Structure and Examples of Esters of Organophosphorus Acids

Structure	Ester	Example
$(RO)_3P=O$	trialkyl phosphate	tri- <i>n</i> -butyl phosphate (TBP)
$R(RO)_2P=O$	dialkyl alkylphosphonate	dibutyl butylphosphonate (TBBP)
$R_2(RO)P=O$	alkyl dialkylphosphinate	butyl dibutylphosphinate
$R_3P=O$	trialkylphosphine oxide	tri- <i>n</i> -octylphosphine oxide

		(TOPO)
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Solvent Extraction-I

The first three members of this group are esters of phosphoric, phosphonic and phosphinic acids and the last member is a derivative of phosphine. The extraction mechanism for all of them is mainly the same. It is the oxygen of the phosphoryl group which is responsible for the coordination bond formed with the metal. In the case of esters, there is a possibility of formation of more than one coordination bond through other oxygen atoms. The inter or intramolecular bifunctional complex, thus, formed exhibits different extraction and stripping rates. The solubility of these extractants in water is very much in the predictable order: phosphates < phosphonates < posphinates < phosphine oxide. This sequence is in line with the increasing polarity of the phosphoryl group in the molecule. The solubility decreases with the increasing chain length.

The high extraction power of neutral organophosphorus esters has been demonstrated for a large number of metal salts. The nitrates and chlorides have received the most attention. The extraction from sulphates and phosphate solution is generally poor. The extraction of metals from halide solutions other than chlorides follows closely that of the latter. The extraction by these reagents involves the solvation of the electrically neutral metal salts formed by their depressed ionization in the aqueous solution. The reagent functions satisfactorily in the presence of high concentration of salting out electrolytes in the aqueous phase. Thus, the extent of extraction will depend on the degree of formation of the extractable species and the solvation number of the metal. There is a strong competition between the extractants molecules and water to occupy the coordination sites of the metal ion. This will depend upon the type and the structure of the neutral organophosphorus extractants. Nearly all metal extractions are increased as the number of direct C – P linkages in the series increase, *i.e.*, in phosphate < phosphonate < phosphinate < phosphine oxide. The salting out effect observed in the case of these extractants is more or less on similar lines as with solvating molecule with carbon oxygen bond. A number of salting out agents have been investigated for their capacity to draw the metal to the organic phase. The extraction increases as the radius of the cation of the salting out agent decreases in the alkali and alkaline earth series. The extraction increases with the increasing charge on the cation. The effect of the diluent on the variation of the distribution ratio of the metal ion can be summarized as under:

- i) At low metal loadings, the distribution ratio is generally higher in aromatic and aliphatic hydrocarbons than in ethers and ketones.
- ii) Especially low values are obtained in polar alcohols and chloroform.
- iii) It still depends frequently on the aqueous phase conditions whether the unsubstituted aromatic or aliphatic hydrocarbons will show higher distribution ratio.

The most extensively used reagents of this class are tributyl phosphate, tributylphosphine oxide, trioctyl phosphine oxide (TOPO). In the Nineteen eighties, the American Cyanamid introduced a new extractant Cyanex 923 in the market. It is a mixture of four trialkylphosphine oxides and has shown a great potential for metal ion separations.

At this concluding stages of the discussion on the extraction by solvating molecules, it may be important to make a brief comment on some other extractants with phosphorus – sulphur bond. This group contains sulphur analogues of the previous group *e.g.* alkylthiophosphoric, alkylthiophosphonic and alkylthiophosphinic esters as well as phosphine sulphides like the Cyanamid reagent, trisobutylphosphine sulphide (TIBPS), Cyanex 471 X. The importance of having analogues of oxyphosphorus reagents with sulphur, is of course, the difference between the oxygen and sulphur affinities for metals. The extractants with oxygen donor are classified as hard bases

while with sulphur donor are soft bases. It is, therefore, expected that hard bases will associate with hard acids and soft bases with soft acids. For metal ions like Fe^{3+} , La^{3+} , Th^{4+} etc. (hard acids), extractants with oxygen donors are a better choice than a nitrogen or sulphur donor. Metals such as Cu^{2+} , Ag^+ , Cd^{2+} , Pd^{2+} , etc. are soft acids, hence, prefer sulphur donor extractants. Cyanex 471 X has been successfully developed as an extractant for Ag^+ , Pd^{2+} and Hg^{2+} .

Another class of extractants which could be of considerable interest is the amide type extractant (A 101). It is a weakly basic extractant with carbonyl being the functional group. The reagent has been used for the separation of Nb and Ta and Mo and Re. Subsequent to development of A 101 in China another reagent N 503, which is *N, N* – dimethylheptyl acetamide has been used for the separation of Nb and Ta.

2.3.4 Extraction by Ion Pair Formation

This primarily involves the extraction of a species formed due to an interaction between an anionic metal species in the aqueous phase and the cation furnished by an organic base. The strongly basic high molecular weight amines (HMWA) are able to extract the metals by forming the ion pairs with anionic metal complexes. The other extractable pairs formed by quaternary phosphonium, arsonium, phenylonium bases are also included under this category. Some of the examples of different types are $\text{R}_3\text{NH}^+\text{FeCl}_4^-$, $(\text{R}_4\text{N}^+)_2.\text{ZnCl}_4^{2-}$, $(\text{R}_3\text{NH}^+)_2.(\text{Cd}(\text{C}_2\text{O}_4)_2)^{2-}$ and $(\text{C}_6\text{H}_5)_4\text{As}^+.\text{MnO}_4^-$. Under this class of extractants, high molecular weight amines have received maximum attention.



The amine salt can undergo anion exchange with anion (B^-) in the aqueous phase, B^- can be simple or complex anion like Cl^- , CrO_4^{2-} or an anionic metal complex like FeCl_4^- , HgCl_4^{2-} and ZnCl_4^{2-} . The order of preference in the organic amine solution is similar to that anion-exchange resins, $\text{Cl}^- > \text{NO}_3^- > \text{HSO}_4^- > \text{F}^-$.

The amine salt on treatment with sodium or ammonium hydroxide can be reverted to amine. The popularity of these extractants probably stems from the fact that the extraction is mainly through ion exchange and thus, they combine some of the advantages of both solvent extraction and ion exchange. They are popularly known as *liquid anion exchangers*.

Generally speaking, liquid ion exchange involves the selective transfer of a solute, an anion or anionic metal complex, from aqueous to organic phase reversibly. The extent of extraction of the metal directly depends on the ease of formation of anionic metal complexes in the aqueous phase. That is to say, aqueous phase parameters like concentration of ligand, hydrogen and metal ion, charge on the complex metal ion involved, the presence of other complexing or competing ion and temperature will not only affect the formation of the extractable species but also, in turn, have a significant effect on the extent of extraction. Furthermore, such ion association systems are also sensitive to organic phase parameters like the nature, structure and chain length of the amine, its concentration and the type of the organic solvent used as diluent. Each of these factors can change the course of the overall extraction process of a metallic species unpredictably. It is apparently a general phenomenon that the extraction power of alkylamines increases from primary to secondary to tertiary to quaternary amines. An exception to this rule is the extraction by bulky amines where steric factors generally affect the basicity of amines, which in turn affects the stability and

The discovery of HMWA as extractants dates back to 1948 when Smith and Page observed that their solution in organic solvents can extract acids from the aqueous phase. Subsequently, the great potential of these reagents for the extraction of anions and anionic metal complex was realized.

polarity of their salts. Due to steric factors, change in the sequence of extractability of amine salts is also observed in the case of anionic carboxylato complexes.

Several efforts have been made to correlate the nature of diluent with extraction power of an amine. As a matter of fact, the lack of correlation between the distribution ratio of a metal and physical characteristics of a solvent such as dipole moment and dielectric constant have led to the conclusion that the amine-diluent combination rather than amine alone should be considered as the effective extractant. There are some limitations in using these reagents as extractants. The most usually encountered is the formation of emulsion. This problem can be effectively tackled by choosing a low surface active extractant and a proper diluent. Control of conditions like temperature, pretreating acid, amount of free acid in aqueous phase and the volume of the two competing phases help to minimize this problem. In some cases, spectacular reduction in emulsion formation has been observed by the addition of 2-3 % higher alcohols.

The existing literature on the extraction behavior of high molecular weight amines reveals that mostly the extraction of metals from mineral acids such as hydrochloric, nitric and sulphuric acid has been investigated. In particular, the extraction of anionic chloro complexes has received maximum attention. Some interest has also been generated in extending the use of HMWA for the extraction of various metal ions from aqueous carboxylic acids. Some work has been reported on the malonate, tartrate, citrate and oxalate complexes.

High molecular weight amines are basically aliphatic primary, secondary, tertiary amines and quaternary ammonium salts with molecular weight in the range 250 to 600. Many of the extractants of this category which are commercially available, are primary amines – Primene JMT, Primene 81R; secondary amines Amberlite LA- 1, Amberlite LA- 2, Adogen- 283; tertiary amines – Alamine 336, Adogen 364, Adogen 382, and quaternary amines – Aliquat 336, Adogen 464.

Some quaternary phosphonium and arsonium compounds have found extensive use in the extraction of simple acid radicals, oxyanions of metals, anionic inorganic coordination complexes involving organic ligands. These extractants behave as strong electrolytes and are usually water soluble. The metal complexes with a well-known stoichiometry are formed when reagent is added to a neutral or weakly acidic solution and they are transformed into a slightly polar organic solvent such as chloroform or methyl isobutyl ketone. Apart from the effect of anion on the extractability of the ion pair, the size of the organic cation bears markedly on the extractive property. Another property which prominently affects the extraction is the nature of the diluent. A very typical example is the extraction of MnO_4^- or ReO_4^- in chloroform formed by tetraphenylarsonium chloride. The extracting Mn(VII) species is proposed to be $(\text{C}_6\text{H}_5)_4\text{As}^+.\text{MnO}_4^-$. Some dimerization of the monomer in the organic phase also takes place.

2.3.5 Extraction by Crown Ethers

A new class of reagents, namely the crown compounds, are fast emerging as potential extractants. Initially, the importance of these compounds was highlighted because of the fact that they are good extractants for alkaline and alkali earth metals and thus, the separation of these metals by solvent extraction became possible. These are multidentate macrocyclic ligands with O, N or S as donor atoms. The binding of the metal is determined by the compatibility of the cavity size of the crown compound and the radius of the metal ion.

The first few compounds of this series were crown ethers which consist of cyclic

($-\text{CH}_2-\text{O}-\text{CH}_2-$) groups to form a cyclic compound having 12-24 ring oxygen atoms. Following the synthesis of crown ethers, several new compounds have been prepared and investigated. These compounds differ in properties owing to the differences in ring atoms and the number of donor atoms. The variation in the ring size leads to the changes in the size and hence, the size of the ion which can be accommodated. By introduction of the linking group, the donor strength of the oxygen can be changed. Also, by introducing S or N atoms instead of oxygen, crown compound of different specificity are synthesized. The introduction of oligoether chains in the classical crown ethers leads to the formation of a new bicyclic type of ligands. These ligands are known as *cryptands* and find considerable application in analytical chemistry. It may be worth mentioning that the discovery of macrocyclic polyether was a chance but the synthesis of macrobicyclic polyethers is carried out in a planned way.

The crown ethers contain a central varying size hydrophilic cavity which is ringed with electronegative binding polyether oxygen atoms. These oxygen atoms are surrounded by a collar of $-\text{CH}_2$ groups imparting it a hydrophobicity. They complex with the cation converting it into a lipophilic species which extracts in the organic solvent. The complexes are composed of two ionic species held together by electrostatic attraction.

The selectivity in extraction is generally explained by the compatibility between cation and hole or cavity size of the crown ether. But the ion-in-the hole model has certain limitations. As the number of ring atoms increases the macrocyclic flexibility increases and it may be difficult to define the cavity diameter. The attachment of certain groups affects the donor strength of the oxygen atoms. Also, the nature of the donor atom plays an important role on selectivity. The extent of extraction is affected by the nature and concentration of the crown ether and the counter ion. The nature of the diluent also affects the extractant. Generally, it is preferred to use the solvents of high dielectric constants. Moreover, the solvent should have high solubility for the crown ethers. The complexation with ethers transforms a relatively small alkaline earth metal ions into a voluminous organic cation which is more soluble in organic media. As a consequence of this, solubilization of inorganic salt of low polarity results. With anions of highly localized negative charge, the solubilization may be difficult. Therefore, the salts containing hard anions like fluoride or hydroxide can be solubilized with difficulty in solvents like benzene. But soft anions like picrate, iodide or tetraphenyl borate are very suitable. The extraction of metals by crown ethers in the presence of picrates or dye stuffs has the added advantage of employing it as a colorimetric method. However, the cost of crown ethers has been a deterrent for their application as extractants on a large scale.

SAQ 4

A certain metal complex, $\text{MX}_n \cdot 2\text{H}_2\text{O}$, formed with the chelating agent HX is poorly extracted in a nonpolar solvent but its extraction is favoured in a polar solvent. Explain why?

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SAQ 5

What are the two basic approaches adopted to achieve selectivity in the metal ion extraction by chelating agents?

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SAQ 6

Solvent Extraction-I

Why do we generally prefer an alkylphosphorus acid for the extraction of a metal ion over a carboxylic acid?

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SAQ 7

The extraction of uranyl nitrate in a solvating extractant is enhanced by the addition of equal amounts of LiNO_3 , NaNO_3 and CsNO_3 to the aqueous phase. Arrange these alkali metal nitrates in their capability to enhance the extraction. Give reasons also.

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SAQ 8

Generally, what is the order of extraction power of high molecular weight amines? Give a possible reason if a deviation in the sequence is observed.

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SAQ 9

What is the common reason for the selectivity in the extraction by crown ethers?

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2.4 DILUENTS AND MODIFIERS

In the previous section, we have seen that different categories of extractants are available and the range is very broad. The different extractants have their own advantages and limitations. One particular point which should be kept in mind is that the majority of extractants are either solids or their viscosities are too much high for their direct use in the solvent extraction. It is, therefore, a common practice to dissolve them in a suitable solvent which is known as **diluent**. Thus, the diluent refers to the organic liquid in which the extractant is dissolved to form the solvent. In majority of cases, the diluent comprises the major portion of the solvent. During the solvent extraction, sometimes the formation of a third phase takes place and to avoid this, a modifier may have to be added to the solvent system. In view of the importance of diluents and modifiers to make an extractant system workable, it may be essential to discuss their characteristics. Proper selection of diluent and modifier can be as important as selecting the extractant because of their physical and chemical effects on the process. Also, they may influence the economics of the process to a great extent.

2.4.1 Diluents

The reasons for using a diluent are new clear to you and they can be enumerated as under:

- i) They are mainly used to decrease the viscosity of the extractant. Generally, the extractants used for commercial solvent extraction processes are viscous liquids and they cannot be used as such.
- ii) Depending upon the metal to be loaded, an appropriate concentration of the extractant is to be used. Thus, the diluent provides the solvent for preparing the solution of the extractant of the desired concentration.
- iii) Another role of the diluent is to decrease the emulsion formation tendencies. A number of extractants are surface active agents and at high concentration, tend to emulsify specially at high agitations.
- iv) The diluent improves the dispersion and coalescence properties of the solvent. These are two very important aspects of a solvent extraction process.

After having seen the various roles of diluents, it may be easy to define their essential characteristics which are listed below:

- i) It should be mutually soluble with an extractant or modifier.
- ii) It should have high solvency for the extracted metal species; thus, providing a high metal loading capacity to the solvent and minimizing the problem of third phase formation.
- iii) It should have low volatility and a high flash point. This minimizes the losses due to volatility and fire hazards.
- iv) It should be in its own right insoluble in the aqueous phase.
- v) It should have low surface tension.
- vi) It should be cheap and readily available in large quantities.

The choice of diluents ranges from simple molecules to complex mixtures. They are both aliphatic and aromatic. The simple molecules that are routinely used at the laboratory scale are *n*-hexane, cyclohexane, benzene, toluene, xylene, chloroform, carbon tetrachloride and chlorobenzene. The use of some of these is restricted due to environmental or toxicological considerations. The cost can also be a deterrent.

One of the important cheap diluents for industrial use is kerosene particularly its 160⁰- 200⁰ C fraction. For commercial applications, particularly in hydrometallurgy, a range of diluents has been developed by major oil companies. In general, these diluents contain a mixture of paraffinic, aromatic and naphthenic hydrocarbons. The Excaid range from Exxon (Escaid 100, Escaid 110, Escaid 350), MSB 210, Shell 140, Shell R from Shell and Chevron Ion Exchange solvents speak of the variety of commercial diluents available.

In the earlier section, we have noticed that the diluents affect the extractability of an extractant to a great extent. It has become apparent that diluents are not as inert, as they might appear to be. Various workers have tried to correlate different properties of the diluents with the extraction efficiency of a system. The mechanism of the role of the diluent seems to be complex. One property of the diluent which has received special attention is its polar nature. In the extraction of rare earth nitrates with Aliquat 336, a decrease in the extraction of the metals with the polar nature of the diluent was observed. Similar effect was noticed on the extraction of anionic chloro species in tertiary amines. From different observations, one may conclude that the diluent affects the solvation of the extractant and hence, its extractive properties. Interaction between the diluent and the extractant lowers the free concentration. The effect of dielectric constant of the diluent in some other systems was observed and no straight correlation could be observed. It is difficult to ascribe the change in extraction solely to the dielectric constant. Other effects such as impurities, degree of aromaticity, solubility of diluent in water, etc. may also be involved.

One thing clearly emerges out of the above discussion that the choice of a diluent for a particular process cannot be predicted, a priori. A diluent is usually picked up on the basis of the experimental data. Diluents are usually produced on industrial scale resulting in some variations in their composition. These variations and the presence of impurities may alter the behaviour of diluents.

2.4.2 Modifiers

One of the very serious problems in solvent extraction is the formation of a third phase *i.e.*, two organic phases. Most of the available evidence on third phase formation suggests that it is essentially a solubility problem. Invariably, it is the organic phase that splits into two phases to yield a metal complex-rich phase at the aqueous interface and a diluent rich phase above. In most of the cases, the problem is overcome by the addition of a third component known as a *modifier*. This helps to restore the organic phases to one. Such modifiers are invariably solvating agents like isodecanol, 2-ethylhexanol, *p*-nonyl phenol and TBP.

The third phases are of a density intermediate between that of the solvent phase and the aqueous phase and result from the effects of the diluent on the heterogeneous equilibrium associated with the system. Based on several observations, some conclusion can be drawn on the third phase formation.

- i) The formation of third phase is temperature dependent. Usually increase in temperature will result in the disappearance of third phase.
- ii) The appearance of the third phase is more common with aliphatic diluents.
- iii) In the case of high molecular weight amine extractants, the tendency for third phase formation with different anions follows the order $\text{NO}_3^- < \text{Cl}^- < \text{SO}_4^{2-}$. A similar sequence is observed with DEHPA.

Addition of a modifier helps to overcome third phase and emulsion formation tendencies in a solvent system. As is with diluents and extractants, the modifier should be very soluble in the organic phase, insoluble in the aqueous phase, be readily available and cheap. Usually, the amount of a modifier used is 2-5 volume percent. In some cases, where the extractant concentration is high, the requirement may be as high as 20% or more.

Experimental evidences suggest that a modifier like a diluent cannot be selected arbitrarily. The selection for a particular system is to be made on the basis of experimental data. The amount of a modifier required is also to be determined experimentally. One point which should be borne in mind is that the effect of a modifier and a diluent in the extraction stage differs greatly from their effects in scrubbing and stripping stages, both on mass transfer and kinetics, and also on phase disengagement and solvent entertainment.

SAQ 10

What are the common requirements for both the diluent and the modifier?

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2.5 SUMMARY

Solvent extraction is a popular separation technique because of its convenience, speed and a favourable economics. This versatile technique can be safely extended from

micro levels to macro concentrations. Some of the important theoretical and practical aspects of the technique have been presented in Units 2 and 3. In Unit 2, the law governing the partitioning of chemical species, its limitations and the different terms used to explain distribution have been discussed. The focus of discussion was on the extraction of metal ions. A classification of different types of extractants used for metal ions has also been described. The characteristics, advantages and the problems faced in the use of different categories of extractants have been highlighted.

The role of diluents and modifiers in solvent extraction system mainly depends upon the composition of the aqueous and organic phases. Finally, the characteristics of diluents and modifiers have been explained.

2.6 TERMINAL QUESTIONS

1. Derive an expression for the distribution of acetic acid between water and benzene in terms of its dissociation and polymerization constants. Also analyze the dependence of distribution on different variables.
2. The distribution coefficient of a certain weak acid between benzene and water is 100. Calculate the distribution ratio for this acid at $pH = 7.00$ of the aqueous phase. $K_a = 1.00 \times 10^{-6}$ for the acid in the aqueous phase and the acid remains as a monomer in the benzene phase.
3. The distribution ratio for a solute X in water-carbon tetrachloride is 10. Calculate the percentage of solute extracted from 50 mL of water by 100 mL of carbon tetrachloride where (a) the total volume of CCl_4 is used at one time and (b) the 100 mL of CCl_4 is divided into five 20 mL portions and employed one after the other.
4. Highlight the differences between chelating agents and acidic organophosphorus esters, carboxylic acids as metal ion extractants.
5. High molecular weight amines extract the metal ions by a mechanism which is similar to their uptake by solid anion exchangers. In what respects, extraction in high molecular weight amines scores over sorption of metal ions by solid anion exchangers?
6. What are the essential characteristics of a diluent?

2.7 ANSWERS

Self Assessment Questions

1. The distribution coefficient, thermodynamic distribution coefficient and distribution ratio will be the same when
 - i) The activity coefficient of the distribution species in the two phases approaches unity, and
 - ii) The distribution species does not undergo any interaction, i.e. association or dissociation in either of the phases
2. $\% E = \frac{100D}{D + V_a / V_o}$

$$\text{Percent of Iodine extracted} = \frac{100 \times 132}{132 + 100/50}$$

$$\begin{aligned}
 &= \frac{1320.0}{134} \\
 &= 98.51\%
 \end{aligned}$$

Iodine remaining in the aqueous phase = 1.49%

$$3. \quad 99 = \frac{100D_A}{D_A + 1}$$

Distribution ratio of A (D_A) = 99

Likewise distribution ratio of (D_B) = 0.01

$$\beta = \frac{D_A}{D_B} = 99/0.01 = 9900 \approx 10^4$$

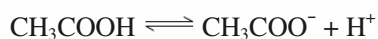
4. Polar solvent can replace the water molecules in the complex; thus, making the species more organic phase loving. Non-polar solvent cannot replace the water molecules.
5. The two approaches generally adopted are
 - i) by adjusting the pH of the aqueous phase, and
 - ii) by adding appropriate masking agent in the aqueous phase.
6. The alkylphosphorus acids are preferred over carboxylic acid because of the following reasons:
 - i) Carboxylic acids have more aqueous solubility and therefore, suffer from the drawback of extractant loss,
 - ii) Carboxylic acids are more surface active and hence, emulsion formation tendency is more pronounced, and
 - iii) Generally, the kinetics of extraction is slower in the case of carboxylic acids than in alkylphosphorus acids.
7. The capability of enhancing the extraction follows the order
 $LiNO_3 > NaNO_3 > CsNO_3$
 Li^+ being the smallest has the highest hydration tendency, followed by Na^+ and then Cs^+ . Because of hydration tendency, the water molecules in the aqueous phase may not be available to compete; thus, resulting into enhancement in the extraction in the organic phase.
8. Generally, the order of extraction power of HMWA is primary < secondary < tertiary < quaternary. The exception is in the case of bulky amines where the basicity is affected. Steric factor may also cause change in sequence in the case of carboxylate complexes.
9. The selectivity in extraction by crown ethers is generally achieved by the compatibility between the cation and the hole or cavity size of crown ether.
10. The common requirements of a diluent and modifier are;
 - i) They should be mutually soluble and should have good miscibility with the extractant.
 - ii) They should have negligible aqueous solubility.

2.8 TERMINAL QUESTIONS

1. The distribution of acetic acid in water and benzene can be described as under
 $(\text{CH}_3\text{COOH})_a \rightleftharpoons (\text{CH}_3\text{COOH})_o$

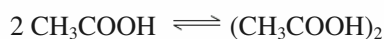
$$K_D = \frac{[\text{CH}_3\text{COOH}]_o}{[\text{CH}_3\text{COOH}]_a} \quad \text{o, and a refer to benzene and aqueous phases, respectively.}$$

However, acetic acid dissociates in the aqueous phase



$$K_a = \frac{[\text{CH}_3\text{COO}^-] \times [\text{H}^+]}{[\text{CH}_3\text{COOH}]}$$

and forms a dimer in benzene.



$$K_D = \frac{[(\text{CH}_3\text{COOH})_2]_o}{[\text{CH}_3\text{COOH}]_a^2}$$

The overall distribution of acetic acid, described by Distribution ratio (D), is given by

$$\begin{aligned} D &= \frac{\text{Total conc. of acetic acid in its different forms in organic phase}}{\text{Total conc. of acetic acid in its different forms in aqueous phase}} \\ &= \frac{[\text{CH}_3\text{COOH}]_o + 2[(\text{CH}_3\text{COOH})_2]_o}{[\text{CH}_3\text{COOH}]_a + [\text{CH}_3\text{COO}^-]_a} \end{aligned}$$

Incorporating the equilibrium constants given above

$$D = \frac{K_D[1 + 2 K_D(\text{CH}_3\text{COOH})]_o}{1 + K_a[\text{H}^+]}$$

This shows that the distribution of acetic acid varies as a function of the pH and acid concentration in organic phase.

$$2. \quad D = \frac{100(1.00 \times 10^{-7})}{(1.00 \times 10^{-7}) + (1.00 \times 10^{-6})} = 9.09$$

$$\begin{aligned} 3. \quad a) \quad \text{Fraction remaining in the aqueous phase} &= \frac{V_a}{D V_o + V_a} \\ &= \frac{50}{(100)(10) + 50} \\ &= 0.04762 \end{aligned}$$

$$\begin{aligned} \text{Fraction transferred to the organic phase} &= 1 - 0.04762 \\ &= 0.9524 \end{aligned}$$

$$\% \text{ Extracted} = 95.24$$

$$\begin{aligned}
 \text{b) Fraction remaining in the aqueous phase} &= \left(\frac{50}{50 + (10)(20)} \right)^5 \\
 &= 0.00032 \\
 \text{Fraction transfer to the organic phase} &= 1 - 0.00032 \\
 &= 0.9997 \\
 \% \text{ Extracted} &= 99.97
 \end{aligned}$$

4. See sub-Sec. 2.3.2 i) – iii)
5. High molecular weight amines score over solid anion exchanger in the following respects:
 - i) The extraction process is much faster than the sorption – desorption on ion exchangers.
 - ii) HMWA – diluent systems offers a wider choice of amines and diluent than that is available with solid anion exchanger. Therefore, several of the separations which are difficult to achieve on solid anion exchangers can be easily accomplished by extraction in an appropriate high molecular weight amine-diluent system.
6. See sub-Sec. 2.4.1

2.9 FURTHER READINGS

1. Solvent Extraction in Analytical Chemistry, By G. H. Morrison and H. Freiser, John Wiley & Sons Inc.
2. Ion Exchange and Solvent Extraction of Metal Complexes, By Y. Marcus and A. S. Kertes, Wiley – Interscience.
3. Solvent Extraction : Principles and Applications to Process Metallurgy Part I, By G. M. Ritcey and A. W. Ashbrook, Elsevier Science Publishers B. V.
4. Solvent Extraction of Metals, By A. K. De, R. A. Chalmers, and S. M. Khopkar, Van Nostrand-Reinhold Publication.
5. Solvent Extraction of Metal Chelates, By J. Sary, Pergamon Press.